

TASK 6: MANAGING AND DEPLOYING AZURE RESOURCES

Chapter 11

Step 1: Azure Resource Manager (ARM)

Azure Resource Manager (ARM) is the deployment and management service in Azure. It allows BCX to create, update, and manage resources in a consistent and organised way. ARM supports features such as resource grouping, role-based access control, tagging, and locks. It also enables batch deployment and automation of resources, ensuring consistency across environments.

Step 2: ARM Template Structure

```
{  
  "$schema": "https://schema.management.azure.com/schemas/2019-04-  
01/deploymentTemplate.json#",  
  "contentVersion": "1.0.0.0",  
  "parameters": {},  
  "variables": {},  
  "resources": [],  
  "outputs": {}  
}
```

- \$schema: Defines the template structure and version
- contentVersion: Version of the template
- parameters: Values passed into the template
- variables: Reusable values inside the template
- resources: The Azure resources to be deployed
- outputs: Values returned after deployment

Step 3: Simple ARM Template (Storage Account)

```
{  
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",  
  "contentVersion": "1.0.0.0",  
  "resources": [  
    {  
      "type": "Microsoft.Storage/storageAccounts",  
      "apiVersion": "2021-04-01",  
      "name": "stbcxsiyabonga",  
      "location": "South Africa North",  
      "sku": {  
        "name": "Standard_LRS"  
      },  
      "kind": "StorageV2",  
      "properties": {}  
    }  
  ]  
}
```

This template deploys a storage account in Azure using Infrastructure as Code. It ensures consistent and repeatable deployments without manual configuration.

Step 4: Deployment Modes

Azure supports two deployment modes:

Incremental Mode (default): Adds or updates resources without deleting existing ones.

Complete Mode: Deletes any resources not defined in the template.

Incremental mode is safer, while complete mode can be risky because it may remove important resources unintentionally.

Step 5: Deployment Tools Comparison

Tool	When to Use
Azure Portal	Easy manual deployment using GUI
Azure CLI	Automation and scripting (cross-platform)
Azure PowerShell	Advanced scripting using PowerShell
Visual Studio / VS Code	Development and Infrastructure as Code

Step 6: Azure CLI Example

```
az deployment group create \  
  --resource-group RG-BCX-Siyabonga \  
  --template-file template.json
```

This command deploys resources using an ARM template into the specified resource group. It automates resource creation and ensures consistency.

Step 7: Azure Arc (Hybrid Management)

Azure Arc allows BCX to manage on-premises, multi-cloud, and hybrid resources from the Azure portal. It extends Azure management capabilities to servers and services outside Azure, enabling centralised control and monitoring.

Step 8: Azure App Service (PaaS)

Azure App Service is a Platform as a Service (PaaS) offering that allows BCX to host web applications without managing infrastructure. It provides features such as automatic scaling, load balancing, custom domains, SSL support, and compatibility with both Windows and Linux environments.